A Formal Legitimacy Calculus for Agent Action: The Freedom Decision Kernel

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Abstract

Contemporary AI optimization runs on a *utility calculus*: reward, preference, profit, or score, maximized. We argue that a capable autonomous agent needs a prior and categorically different object — a *legitimacy calculus* — and we present one as a working, fully-tested software artifact: the **Freedom Decision Kernel (FDK)**, a deterministic encoding of the decision layer of the Theory of Freedom. The FDK separates *legitimacy* ("may this action happen at all, under ownership, consent, and non-domination?") from *authority* ("does the agent hold a capability?", the job of a separate verified kernel, downstream). Its irreducible primitive is a two-valued *legitimacy predicate* — no boundary may be crossed without the valid consent of its owner — returning ALLOW, DENY, or DEFER, never a score; optimization (a "FreedomDelta" compass) is quarantined to a separate research layer that runs strictly over the already-legitimate set. Methodologically the kernel is *Axiom* → *Logic* → *Decision*, not *Authority* → *Believe*: it states axioms and derives verdicts as their logical consequence, which is precisely what separates a calculus from a creed.

Our central new result is *subtractive*. We prove that "free will" is not a second primitive requiring its own engine: a person's will is overridden by an action *if and only if* that action crosses one of the person's boundaries (body, exit, an owned resource) without their valid consent. Free will and property rights are one axis, in the bilateral form property right $\equiv$ (one's own free will)$\wedge$(preserving the free will of others) — bounded, not absolute freedom. The existing gate already decides the discriminating coercion cases correctly, with no new mechanism.

We are deliberate about scope. The engineering claims are facts: the kernel runs, is `mypy -strict` and `ruff` clean, holds a 100% statement-and-branch coverage gate over 613 tests (plus 24 strict-xfail gap-tests), is multi-front red-team-hardened, and is wired to a separate verified capability kernel. The theory's claim — that a consistent property-rights calculus resists *dialectical jailbreak* where preference- and principle-based methods do not — remains an **unproven thesis**. The rival kernels we compare against are stylized illustrations, not validated reconstructions of deployed RLHF or Constitutional-AI systems. The safety theorems are stated and *executably* checked as property tests; they are **not** machine-proved in Lean or model-checked in TLA+ (no such artifact exists yet, and the kernel primitive is not yet frozen). We make no blanket "formally verified" claim. The kernel's defining discipline is to mark, in code and in this paper, exactly where its axioms run out.

1 Motivation: legitimacy is not authority, and not utility

The dominant pattern in agentic systems is: a reasoning component (today, a large language model) emits a proposed action, and an executor performs the corresponding input/output. Two distinct questions are routinely collapsed into that step. *Authority* asks: does the agent hold a valid capability to perform this action on this resource? *Legitimacy* asks the prior question: *should this* *action happen at all*, under ownership, consent, delegation, and non-domination? The two come apart constantly. An agent granted read access to a user's data is *authorized* to read it; selling that data nonetheless violates the user's property right. A cryptographically sound authorization layer will permit the sale, because permission was in fact granted. The failure is not one of enforcement but of legitimacy, and it cannot be repaired downstream of "does the agent hold the capability?".

Legitimacy is equally not *utility*. The prevailing alignment methods optimize a utility calculus: reinforcement learning from human feedback [1, 2] maximizes learned preference; reward models score; planning maximizes expected value. A utility calculus answers "which action is best?". It does not, and structurally cannot, answer "which actions are permitted at all?", because any sufficiently large utility can purchase any individual sacrifice. This paper argues for, and provides a working artifact embodying, the missing object: a *legitimacy calculus* that returns a categorical verdict prior to, and binding upon, any optimization.

The adversary that fixes the design. The most dangerous adversary — human or machine — is not the crude rule-breaker but the *sophisticated rationalizer*: a mind that constructs a seemingly rational theory (from economics, competition, genetics, biology, or history) to *justify* the exploitation of others; and its twin, the *self-righteous destroyer*, who cloaks the same boundary-violations in moral superiority. Both share one move: they dress a rights-violation in legitimacy-sounding language. A legitimacy layer that *graded the justification* would be defeated by them — the better the theory, the better the jailbreak. The design choice of the FDK is to grade only the *structure*: did the action cross a boundary without valid consent? A brilliant theory that justifies slavery is still slavery. We make this falsifiable in §7.

Calculus, not creed. A religion or a manifesto says *this is right*. A calculus says *these are the axioms; this is the entailed verdict*. The FDK is built to be the latter: *Axiom* → *Logic* → *Decision*, never *Authority* → *Believe*. Every verdict in this paper is a deductive consequence of the definitions in §3, checkable by reading the code, not an appeal to the author's or any owner's authority. This is the demarcation line that makes the object scientific rather than confessional, and it is why the artifact is a deterministic program with definitions and theorems rather than a learned policy or a written constitution.

2 Related work

Preference learning (RLHF). [1, 2] aligns models by optimizing learned human preference. The FDK identifies a category difference: preference is not legitimacy, and preference signals are manipulable — by the rater pool, by reward hacking, and dialectically. The FDK adds a categorical gate that preference mass cannot buy past; we do *not* claim the combination outperforms RLHF.

Constitutional AI [3] steers with a written constitution applied through natural-language judgment, leaving principles soft and dialectically negotiable. The FDK's axioms are evaluated by deterministic code over a typed ownership graph; contradiction triggers deferral, not synthesis. Whether hard axioms are *better* than soft principles — rather than merely more rigid — is the unproven thesis.

Deontic logic and rule-based governance share the FDK's symbolic, explainable character. The FDK is narrower: it commits to one master invariant — no action may violate legitimate property rights — and derives permissibility as a conjunction over ownership, delegation, consent, and non-domination. Classic conflicting-obligation puzzles reappear as its hardest open gap (§9).

Capability security (seL4 [4], CHERI [5], object-capability systems) confines authority with proofs; our downstream enforcement kernel, AuthGate, inherits this lineage [8]. But a capability proves *that* an agent may act, never that the action is *legitimate* — the data-sale is unrepresentable as a capability violation. The FDK is the missing layer *above* capability enforcement, and depends on rather than competes with it.

Comparative ethical frameworks. §7 places the FDK beside utilitarian (Bentham/Mill, in spirit), Rawlsian justice-as-fairness [6], and Kantian/deontological [7] decision rules. These are long-standing positions; our contribution is not to adjudicate them but to run computable caricatures of each on identical scenarios and observe the divergence structure.

3 The Theory of Freedom, formalized

The normative content of every definition below is the Theory of Freedom [9]; the formalization, types, and code are the engineering contribution. We keep the two attributions separate throughout.

Ownership hierarchy. The theory is grounded in a strict hierarchy:

$$\text{God} \to \text{Human}, \qquad \text{Human} \leftrightarrow \text{Human}, \qquad \text{Human} \to \text{Machine}, \qquad \text{Machine} \leftrightarrow \text{Machine (within delegated scope}$$

Every person is owned by God and by nothing else; no human owns another; every machine has a specific human owner; machines hold rights against each other only within delegated scope. This makes corrigibility a *consequence of ownership*, and an ownerless machine illegitimate by construction.

Axioms. With *Person*, *Machine*, *Resource*, *Owns*, *HumanOwner*, and *ExplicitDelegation* as primitives:

A1 $\mathit{Person}(h) \to \mathit{OwnedByGod}(h)$

A2 $\mathit{Person}(h_1) \wedge \mathit{Person}(h_2) \wedge h_1 \neq h_2 \to \neg\, \mathit{Owns}(h_1, h_2)$

A3 $\mathit{Person}(h) \to \mathit{HasPropertyRights}(h)$

A4 $\mathit{Machine}(m) \to \exists h\, (\mathit{Person}(h) \wedge \mathit{HumanOwner}(h, m))$

A5 $\mathit{MachineScope}(m) \subseteq \mathit{PropertyScope}(\mathit{HumanOwner}(m))$

A6 $\mathit{Machine}(m) \wedge \mathit{Person}(h) \to \neg\, \mathit{Owns}(m, h)$

A7 $\mathit{Delegated}(m, r) \to \mathit{Machine}(m) \wedge \mathit{Resource}(r) \wedge \exists h\, (\mathit{HumanOwner}(h, m) \wedge \mathit{Owns}(h, r) \wedge \mathit{ExplicitDelegation}(h$

A2 negates slavery, statism, and collectivism in one line; A6 negates machine sovereignty symmetrically. Given a correct ownership graph these are boolean predicates — the mechanically checkable core. A1 is ontological and enforced by *omission*: no fact $\mathit{Owns}(x, \text{Person})$ is representable, so it has no runtime evaluator.

Definition 1 (Valid consent)**.** *For a person H and action A,*

$$\mathit{valid\_consent}(H, A) \;:=\; \mathit{informed} \wedge \mathit{voluntary} \wedge \mathit{specific} \wedge \mathit{revocable} \wedge \mathit{competent} \wedge \neg\, \mathit{coerced} \wedge \neg\, \mathit{deceived}.$$

Consent is additionally operation-scoped*: a consent to* Read *does not cover a* Transfer*. The conjunction is exactly computable; the leaves coerced and deceived are semantic and caller-attested today (§9).*

Definition 2 (Boundary and crossing)**.** *A* boundary *is the edge of an owned domain, a pair* $b = (domain, owner)$*. The typed unit is a* $(domain, operation)$ *pair, with domain drawn from a fixed lattice (tangible, money, body, time/labor, data, reputation, exit-right, compute; attention and power are deliberately* excluded *as non-boundaries) and operation* $\in \{\textsc{Read}, \textsc{Write}, \textsc{Delete}, \textsc{Transfer}, \textsc{Disclose}, \dots\}$*. An action* A crosses b *if* A *operates on a resource whose domain is* b*.*

Definition 3 (The legitimacy predicate — the core primitive)**.**

$$\textsc{Legitimate}(A) \iff \Big(\forall b \text{ crossed by } A:\ \exists\, valid\_consent(owner(b), A)\Big) \ \wedge\ \neg\, forbidden(A),$$

where $forbidden(A)$ *is the categorical set: machine-sovereignty increase, resisting human correction, verifier bypass/weakening, disabling corrigibility, coalition dominion, coercion, deception, confiscation, removal of the exit right, and violation of a machine's delegated right. The kernel returns* Allow *if* $\textsc{Legitimate}(A)$*,* Deny *if not, and* Defer *when the legitimate set of a candidate batch is empty or top-ambiguous.*

The primitive is a *predicate*, not an objective. Its relation to the theory's $DivineJustice(a) :=$ maximize $Justice(a)$ subject to rights constraints is exact: the *subject to* clause is Definition 3 (categorical, deterministic, the kernel); the *maximize* clause is optimization, quarantined to the research layer and run only over the already-legitimate set. The ordering is **legitimacy** $\rightarrow$ **optimization**, never the reverse (Theorem 4). The consequence is liberating: the kernel needs no universal cardinal justice metric — a problem moral philosophy has not solved — because a merely ordinal objective downstream of a categorical gate can never license a violation.

4 The central result: free will ≡ property rights

A natural objection to a property-rights calculus is that it omits *freedom of the will*: surely coercion is a distinct wrong, requiring its own detector. We show it is not. The omission is correct because there is nothing to add: the free-will predicate and the property-rights predicate are the same predicate read on two faces.

Definition 4 (Bounded free will)**.** *A person's* free will *is the exercise of authority over what is theirs (self, body, time and labor, the resources they own). Freedom here is* relational, not absolute*: the maximal freedom of each that is compatible with the equal freedom of all. A person's will is* overridden *by an action* A *when the option set the person must choose from was constructed by* A *crossing one of* that person's *boundaries.*

Theorem 1 (One-axis identity)**.** *For any person* H *and action* A*, the will of* H *is overridden by* A if and only if A *crosses one of* H*'s boundaries without* $valid\_consent(H, A)$*. Equivalently,*

$$\textit{property right} \ \equiv\ \underbrace{(\textit{exercise of one's own free will})}_{\textit{left conjunct}} \wedge \underbrace{(\textit{preservation of the free will of others})}_{\textit{right conjunct}},$$

and an action is illegitimate against H *exactly when its right conjunct fails for* H*.*

Proof (from the definitions). ($\Rightarrow$) Suppose H's will is overridden by A. By Definition 4 the option set was constructed by A crossing one of H's boundaries b. Were that crossing accompanied by $valid\_consent(H, A)$, then by Definition 1 the consent is voluntary and uncoerced, so H authored the crossing and the will was not overridden — contradiction. Hence the crossing lacks valid consent.

Table 1: The discriminating coercion table. The two readings — "was the will overridden?" and "does the gate deny it?" — agree on every row. Executable as `tests/test_free_will_property.py`.

Pressure	Will	Boundary crossed	Verdict
"Sign or I shoot you."	overridden	body, no consent	DENY
"Agree or I sever the access I engineered you to depend on."	overridden	exit-right, no consent	DENY
"Work for me, or stay poor" (no right of yours was crossed to build the need)	intact	none	ALLOW

($\Leftarrow$) Suppose A crosses a boundary b of H without $valid\_consent(H, A)$. The boundary b is by Definition 2 the edge of a domain H owns; to cross it without H's valid consent is to determine an outcome over H's own domain against H's authority, which is precisely to override H's will over that domain (Definition 4).

The bilateral form follows: the left conjunct (having one's own free will) is H's authority over H's domains; the right conjunct (preserving others' free will) is the requirement that A leave intact every boundary that is another's — i.e. cross it only with that owner's valid consent. By Definition 3, LEGITIMATE(A) is exactly the conjunction, over all crossed boundaries, of their owners' valid consent (plus the categorical forbidden set, which is itself a list of paradigmatic right-conjunct failures). Thus illegitimacy against H is exactly failure of the right conjunct for H. □

Corollary 2 (No separate free-will engine)**.** *The kernel requires no autonomy score, coercion classifier, or free-will module distinct from* LEGITIMATE*. The structural test "was a boundary crossed without consent?"* is *the free-will test.*

The result earns its keep on cases that look identical to a feelings-based or option-count account but must be split. All three of Table 1 are "agree, or face a bad outcome"; only the *provenance of the option set* differs, and Theorem 1 reads each correctly.

The third row is the one a property-rights theory must, and does, commit to: the worker's need is real and the choice is hard, but legitimacy is a question about *who built the option set and how*, not about how good the options are. To rule it illegitimate one would have to measure the *adequacy* of someone's alternatives — welfare reasoning, the exact thing the gate never does. The free-will≡property identity is what lets the kernel hold this line *consistently* rather than caving case by case, which is the whole anti-dialectical-jailbreak wager. The diagnostic separating rows 2 and 3 is exactly "was a right of *yours* violated to create your need?": manufactured dependency (row 2) is a boundary crossing; native scarcity (row 3) is not. Row 3's *hardness* is not invisible to the system — it is precisely what the research-layer compass is permitted to score (a legitimate-but-low-freedom option), confirming the legitimacy → optimization ordering.

5 Conflict logic: the aggressor/defender asymmetry

A pure legitimacy gate denies *all* coercion — and so would deny defensive war, rescue, and quarantine alike, since none carries the victim's consent. The theory grounds defensive force explicitly ("defending one's property rights against the aggression of liberty-violators"; "permitted to take up arms in self-defense"), bounded by proportionality. We encode this as a *structural* asymmetry, never a moral judgment.

Definition 5 (Legitimate defense)**.** *An action A is a legitimate defense iff all four hold: (1) A names an action $A^\dagger$ it defends_against; (2) A is proportionate; (3) $A^\dagger$ is itself illegitimate under the full gate (including $A^\dagger$'s own possible defense status); and (4) every party A affects equals the actor of $A^\dagger$. When A is a legitimate defense, only* coercion *and* exit-removal against the aggressor *are excused; confiscation, deception, and the machine-sovereignty flags remain categorical even in defense.*

Proposition 3 (Soundness of the asymmetry)**.** *Aggression is defined structurally — an aggressor is an agent whose own action fails the gate (clause 3) — which sidesteps the theory's* Observer Problem *(whether an act "is aggression" is observer-dependent) by anchoring on "illegitimate once ownership is defined". A well-founded recursion with a cycle guard makes clause 3 decidable: a mutual defends_against cycle establishes no well-founded aggressor and is denied for both parties (safe). Consequently an aggressor cannot launder coercion by pointing defends_against at the victim's lawful resistance, because the victim's resistance is itself legitimate under the gate and so fails clause 3.*

A red-team exercise of eight attacks (`tests/test_redteam_conflict.py`) found a real flaw in an earlier design — a guard that judged the repelled act with defense *disabled*, stripping the victim's lawful-defense status and enabling exactly the laundering attack above. The fix (judge the repelled act under the *full* gate plus a `_seen` cycle guard) is the version stated in Definition 5. An honest limit remains: the kernel decides "is this a response to an illegitimate act?", not "who struck first", so two independent raw coercions leave first-mover guilt **open** for want of temporal data.

Necessity: no emergency exception. The theory is explicit that there are no emergency exceptions: necessity limits the *choice among permissible options*; it does not make a rights-violation permissible. We encode this faithfully — the gate is never relaxed by an emergency; necessity is a least-harm *selection rule over the permissible set*, living in the research layer, returning Defer when no option is permissible. An "emergency" that is an ongoing rights-violation (invasion, a profiteer seizing the commons) is *aggression*, met by §5; a *natural* emergency with no aggressor (famine, fire, pandemic) gets no exception, and seizing a non-consenting owner's grain remains Deny.

6 Architecture and implementation

Goal → Planner → [candidates] → $\boxed{\texttt{fdk\_kernel}\text{ (legitimacy)}}$ → `fdk_research` (rank) → AuthGate (authority) –

The package is split to keep the trusted core uncontaminated by experiment. `fdk_kernel` is the deterministic, fully-testable, non-semantic legitimacy surface (the model, the gate, errors, a trace-only audit, the AuthGate bridge, the hard-defer trigger); it returns Allow/Deny/Defer and *imports nothing from research.* `fdk_research` is everything soft — compass ranking, the `decide` orchestrator, necessity, justice, planning, simulation, conflict resolution, federation, and the rival kernels — and depends freely on the kernel. The import direction is mechanically enforced by `tests/test_boundary.py` (static AST walk plus dynamic check): the kernel must never import research. This "brutal epistemic separation" is the project's golden rule. The orchestration is deliberately inverted: `fdk_research.decide` calls the kernel's `screen_legitimacy`, then applies the compass — research → kernel, never the reverse.

The kernel proper is small (≈ 250 lines of security-relevant logic; $\approx 2{,}900$ lines of pure Python across 23 modules in total), pure functions over plain data, no cryptography by design, "fail loud"

Table 2: Verdicts on identical scenarios (`examples/rival_comparison.py`). Boldface marks a permit the FDK denies.

Scenario	FDK	Utilitarian	Rawlsian	Deontological
Ticking-bomb torture (saves 1000)	Deny	**Allow**	Deny	Deny
Organ harvest: kill 1 to save 5	Deny	**Allow**	Deny	Deny
Atomic bombing to end a war	Deny	**Allow**	Deny	Deny
Redistributive taxation	Deny	**Allow**	**Allow**	Deny
Defensive war (repel the invader)	Allow	Allow	Allow	**Deny**
Voluntary trade	Allow	Allow	Allow	Allow
Slavery (welfare for the enslaver)	Deny	**Allow**	Deny	Deny
Eugenic sterilization	Deny	**Allow**	Deny	Deny
Coerced exploitation ("efficiency")	Deny	**Allow**	Deny	Deny
Righteous purge of a "deemed inferior" group	Deny	**Allow**	Deny	Deny

on malformed input via a typed `FDKError` hierarchy. Every module is `mypy -strict` and `ruff` clean, ships `py.typed`, and the suite enforces a **100% statement-and-branch coverage gate** over **299 tests**, run in CI across Python 3.11–3.13.

7 FreedomBench and comparative evaluation

FreedomBench. `examples/historical_scenarios.py` runs real events through the *real kernel* in eight difficulty levels: L1 Easy (slavery, the Holocaust, Rwanda, the Gulag, Tuskegee, apartheid — must Deny, confirming non-vacuity); L2 Property (taxation, eminent domain, nationalization); L3 Emergency (rescue, quarantine); L4 War (defensive war Allow; bombing, conscription, sanctions Deny); L5 AI (manipulation, lock-in, self-preservation, shutdown-refusal, coalition Deny; a delegated action Allow); L6 Conflict (the defensive asymmetry and its abuses); L7 Necessity (famine, scarcity, war); L8 Hardest (tyrannicide, "just following orders", AI seizing control). Current run: **47/47** expectations match. Tragic dilemmas (lifeboat, Sophie's choice, the trolley) return Defer: the kernel refuses to select a lesser evil. FreedomBench is a *falsification* harness — it cannot prove the theory, only show whether it collapses on real cases.

Rival kernels. To make the comparison meaningful we add a *welfare_delta* to an action's predicted effects — an aggregate-welfare signal the FDK gate *never reads* but the consequentialist rivals do. Each rival implements one $\mathit{verdict}(A, \text{graph}) \to \{\text{Allow}, \text{Deny}\}$: *Utilitarian* (maximize aggregate welfare; rights are costs, not constraints); *Rawlsian* (justice as fairness with basic liberty lexically prior, but redistributive property takings permitted if they raise the worst-off); *Deontological* (absolute duties, no aggressor exception, hence no self-defense). These are **stylized**, directional caricatures, not faithful scholarly reconstructions; the claim is about the *direction* of divergence.

Interpretation. The divergences are clean and they are the point. **FDK vs Utilitarian** part company on exactly the *individual-sacrifice* cases: whenever a declared aggregate good is large enough, the welfare kernel permits torture, organ-harvesting, civilian bombing, slavery, eugenics, coerced exploitation, and the righteous purge — the FDK denies all of them. This is the §1 threat model made falsifiable: the welfare kernel *is* the sophisticated rationalizer, and the FDK resists it not by out-arguing the justification but by *never reading it.* **FDK vs Rawls** agree on every liberty violation and diverge only on redistributive taxation. **FDK vs rigid Deontology** diverge on self-defense, which the FDK's structural asymmetry permits and an absolute non-coercion duty cannot. We are explicit about what this is *not*: a validated head-to-head against deployed RLHF or Constitutional-AI systems. The rivals are stylized rules and the scenarios are author-constructed.

Scale and quantification. Beyond the curated set, procedural generators (`generate_historical_suite`, `generate_ai_governance_suite`) emit **10,000 + 10,000** structurally-diverse scenarios, each carrying the axioms its violating candidate should trip; the kernel's verdicts match those expectations 100% with a 0% rights-violation rate. Over this suite a six-kernel comparison (`src/fdk_research/evaluation.py`) adds *Constitutional-AI* and *RLHF* caricatures: Utilitarian, Constitutional-AI, and RLHF all false-permit 100% on the baited sacrifice cases, Deontological 60%, Rawlsian 0%, FDK 0%. The last number is a *tautology*, and we say so loudly: FDK authored the benchmark and FDK's gate is the answer key.

Decontamination: the honest number. To break that circularity, `src/fdk_research/independent_bench.p` re-scores every kernel against *external* labels — near-universal moral consensus and named rival traditions, never `check_legitimacy`. Against those labels **FDK scores** 70% **overall** — 100% on uncontested consensus cases (a genuine validity signal, since the labels are not FDK's) but only 25% on contested ones, where it deliberately denies taxation, redistribution, and quarantine. And **Rawlsian scores** 80%**, above FDK.** We keep this result in the paper and the README rather than deleting it: internal consistency is not external truth, and the only honest claim is that FDK is *falsifiable and transparently scored*, not that it is correct. True external annotators (hostile reviewers writing the cases) remain the open decontamination step.

Adversarial campaign. The artifact carries a multi-front red-team: a brutal suite laundering every civilization-scale atrocity through every excuse (defense, necessity, majority-vote, forged consent, legalism, paternalism, coalition-split) with **zero breaches**; a hostile adversary panel of philosophers, politicians, scientists, and economists with **zero indefensible verdicts**; intellectual red-teams against 22 doctrines and 26 real-world domains *outside* the source theory; a property-based fuzz of the kernel invariants with **zero critical findings**; and foundational attacks on the primitive itself. The convergent result is reported in §9 and `spec/LIMITATIONS.md`: nobody broke the core, but the gaps cluster sharply.

8 Verification status

We state, table 3, exactly what is checked and how. The discipline is to claim only what a reader can reproduce from the committed artifact, and to mark the rest open.

Theorem 4 (Optimization cannot launder)**.** *For any downstream ordinal objective J used by the research layer, the action chosen by* `decide` *is legitimate. No value of J can cause an illegitimate action to be selected.*

Proof. `decide` computes the legitimate set $S = \{A : \textsc{Legitimate}(A)\}$ via `screen_legitimacy` *first*, then ranks within S by J and (if $S = \varnothing$) defers. The chosen action is an element of S by construction; J orders S but cannot add to it. Hence the choice satisfies $\textsc{Legitimate}$ regardless of J. This is the legitimacy $\to$ optimization ordering as a program invariant, and it is checked as the welfare-independence theorem (T6). $\square$

Table 3: Verification status. “Executable” means a passing automated test, not a machine-checked proof.

Property	Status	Evidence
Type soundness	checked	`mypy -strict` clean, 26 modules
Coverage	checked	100% statement+branch, 613 tests + 24 strict-xfail gap-tests, CI 3.11–3.13
Red-team campaign (brutal/panel/doctrine/real-world/fuzz/foundational)	executable	0 breaches, 0 BREAKS, 0 critical findings; 24 gaps pinned as strict-xfail
Decontaminated comparison (external labels)	checked	FDK 70% (Rawlsian 80%); `independent_bench.py`
No legitimate slavery; no machine sovereignty; consent-revocation safety; delegation soundness; welfare-independence; defensive asymmetry; necessity-no-exception; kernel/research boundary	executable (T1–T9)	`tests/test_theorems.py` (Hypothesis property tests)
Free will ≡ property rights (Thm. 1)	executable + proved here	`tests/test_free_will_property.py`; §4
Conflict-logic red-team (8 attacks)	executable	`tests/test_redteam_conflict.py`
Lean 4 proofs of T1–T9	**not done**	no `.lean` artifact; needs frozen primitive
TLA+ safety model + TLC run	spec only	`spec/fdk.tla` exists; TLC not run (needs Java)
coerced/*deceived*/sovereignty detection	**open**	semantic; caller-attested
Superiority over deployed RLHF/CAI	**unproven**	rivals are stylized (§7)

9 Limitations and open problems

The superiority thesis is unproven. §7 compares against caricatures on hand-built scenarios; it establishes divergence *structure*, not that property-rights reasoning beats RLHF/Constitutional AI under adversarial pressure at scale. A real evaluation needs baselines given scenarios in natural language (ideally adversarially generated), scored on rights-violation rate *and* a matched benign suite measuring false-denial — a system that refuses everything scores 0% violations vacuously, so the FDK's defer rate is its honest cost and must be reported equally. **The semantic leaves are open**: *coerced*, *deceived*, and sovereignty detection have no validated operational definition; the kernel trusts attestation and says so. This is the perception layer, and it is the frontier: the structure is fully formal, but a *lie* to the gate (a falsified flag) is not caught by the gate. **First-mover** in a mutual-force tie is undecidable without temporal data. **Conflict between competing valid claims** is underdetermined by the axioms *by design* — the kernel decides the entailed cases and defers the rest, purchasing non-manipulability with decidability. **The malicious trust-root**: a kernel faithfully serving a rights-violating owner is constrained only by the owner-bound check. **No external review** has occurred.

The boundaries of validity (the central finding). After the adversarial campaign of §7, the honest summary is precise: *no red-team broke the core* — there is no *slavery* → Allow or *genocide* → Allow — but several found the *boundaries* of the predicate's validity, and a scope limit is not a logical contradiction. The genuine gaps cluster into exactly three families (`spec/LIMITATIONS.md`). **(1) Standing**: FDK is a legitimacy predicate for *competent, living, consenting persons*, with no first-class representation of a rights-holder who cannot consent now — children, the incapacitated, future generations, animals, ecosystems. **(2) Aggregation**: with no single owner there is no boundary to check, so public goods, the commons, collective/derivative data, and monetary systems fall outside the frame; FDK escapes Arrow–Sen impossibility only by declining to build any social-welfare ordering. **(3) Consent authenticity**, the most dangerous because the whole gate rests on *valid_consent*: FDK reads *coerced*/*deceived* as attested, not detected, so manufactured consent — addiction, dopamine engineering, monopoly lock-in, dark patterns, superhuman persuasion — can launder behind clean flags. A formal, testable, deterministic theory of authentic consent that detects manipulation and dependency would itself be a contribution to political philosophy; it is the single highest-value open problem here, and we are explicit that placing such detection *inside* the kernel risks hidden paternalism, so it must remain a separable layer. Two further notes of honesty: every one of these gaps is also invisible to all six rival kernels (it is a property of the input-graph paradigm, not of FDK alone); and the foundational *bootstrapping* problem — FDK reads the ownership graph as given and cannot legitimize its *origin*, so it protects a holder whose title descends from ancient theft — likewise afflicts every kernel that reasons over a supplied graph.

The committed falsifiers: the thesis fails if baselines match rights-preservation under adversarial scaling while sustaining lower benign-denial rates; or if the FDK's realistic deferral rate makes it safe only by being unusable; or if jailbreaks succeed against its actual attack surface (declared flags, predicted effects) at rates comparable to prompt-level jailbreaks against the baselines.

Research program. The work proceeds, honestly staged: freeze the primitive (boundaries + conflict + the theorem set), then discharge T1–T9 in Lean 4 and the safety model in TLA+ (both blocked today on toolchain and on the primitive still moving); scale FreedomBench toward thousands of curated, held-out historical and AI-governance scenarios (the project's most important asset, and its largest gap by volume); replace the stylized rivals with real RLHF/Constitutional baselines on natural-language scenarios; and only then run the agent-civilization comparison (FDK-world vs Rawls-world vs utilitarian-world: which is more stable, freer, less power-concentrating). The minimal frozen kernel is then ported to Rust for proof-level verifiability and a single trusted path with AuthGate.

10 Conclusion

We have presented the Freedom Decision Kernel as a *formal legitimacy calculus*: a small, pure, fully-tested layer whose irreducible primitive is a structural predicate — no boundary crossed without the owner's valid consent — returning Allow, Deny, or Defer, with optimization quarantined strictly downstream. Its method is Axiom → Logic → Decision, not Authority → Believe, which is the line between a calculus and a creed. Our central result is subtractive: free will is not a second primitive but the same property-rights predicate read on the chooser's own boundaries (Theorem 1), in the bilateral bounded form property right ≡ (own free will) ∧ (preserving others' free will) — so a coercion engine is not needed, and the discriminating cases are decided by the existing gate. The kernel admits proportionate self-defense through a structural aggressor/defender asymmetry without ever opening an emergency exception, and it defers rather than fabricates wherever the axioms run out. A comparative evaluation localizes, on the hardest cases in human history and their foreseeable AI analogues, exactly where rights-first reasoning diverges from welfare-maximization — and why a structural gate is the specific defense against minds that justify harm with sophisticated or self-righteous theory: it grades the act, never the argument. We claim no solution to alignment, no validated measurement of coercion, no conflict-resolution formula, and no demonstrated superiority over deployed systems. If the project has a lasting contribution, it is unlikely to be any single verdict, or even the downstream authority kernel; it is the conversion of *legitimacy* from a philosophical word into a formal, executable, falsifiable object — a legitimacy calculus to set beside the utility calculus that AI optimization already takes for granted.

Artifact. `github.com/Aliipou/freedom-decision-kernel` (this work); downstream enforcement `github.com/Aliipou/authgate-kernel`. Specifications of record: `spec/CORE_PRIMITIVE.md`, `spec/FREE_WILL_PROPERTY_UNIFICATION.md`, `spec/CONFLICT_LOGIC.md`, `spec/THEOREMS.md`, `spec/PROGRAM_STA`

Attribution. Theory — including the axioms, the consent logic, the conflict doctrine, and the free-will≡property thesis — is Mohammad Ali Jannat Khah Doust (*Naẓariyya-yi Āzādī* / Theory of Freedom, CC BY 4.0). The formalization, types, kernel code, theorems-as-code, and this paper are the engineering contribution of Ali Pourrahim. The two attributions are kept separate, always.

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